



1.9.5 Wrap-Up: Cool Down

1. Write an explanation to a classmate who might have missed today's lesson, in your own words, about copying an angle.

Unit 1, Lesson 10: Segment Bisectors



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships and theorems of lines and angles.

MA.912.GR.5.2 Construct the bisector of a segment or an angle, including the perpendicular bisector of a line segment.



Learning Targets

- ☐ I can construct the bisector of a segment.
- ☐ I use the definition of a midpoint to solve problems.



1.10.1 Warm-Up: Notice and Wonder

In May 2021, the world's longest suspended pedestrian bridge opened in Arouca, Portugal. The bridge runs over 1700 feet across a canyon, nearly 600 feet above the Paiva River.



1. What do you notice?
2. What do you wonder?



1.10.2 Exploration: Segment Bisector

1. Complete the following.

- a. Copy $\overline{TK}$ onto patty paper.



- b. Follow the steps on the copy of $\overline{TK}$.
 - Fold the copy of $\overline{TK}$ that is on patty paper so the endpoints T and K align.
 - Place point M on $\overline{TK}$ where the paper folds.
 - Copy point M onto $\overline{TK}$ above.
 - Draw a line, a line segment, or a ray through point M .
- c. Discuss with your partner what you notice about $\overline{TM}$ and $\overline{MK}$.

The **midpoint** of a segment divides it into two congruent segments.

2. $\overline{WX}$ is shown where point P is the midpoint.



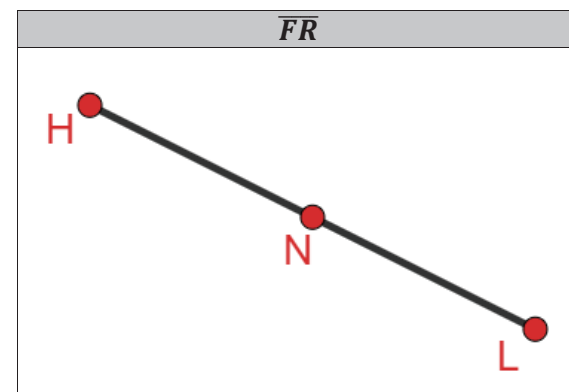
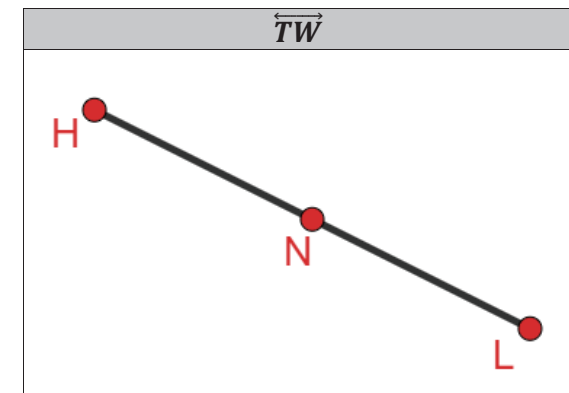
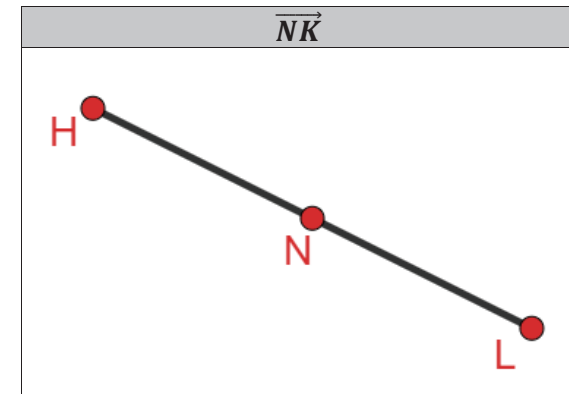
- a. Discuss with your partner what transformations can be used to show that $\overline{WP}$ maps to $\overline{PX}$.
- b. Ask two classmates which transformations they selected. Record their transformation and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Transformation	My Summary of Their Reason	Initials

A **segment bisector** is a line, ray, or line segment that cuts a line segment into two equal parts.

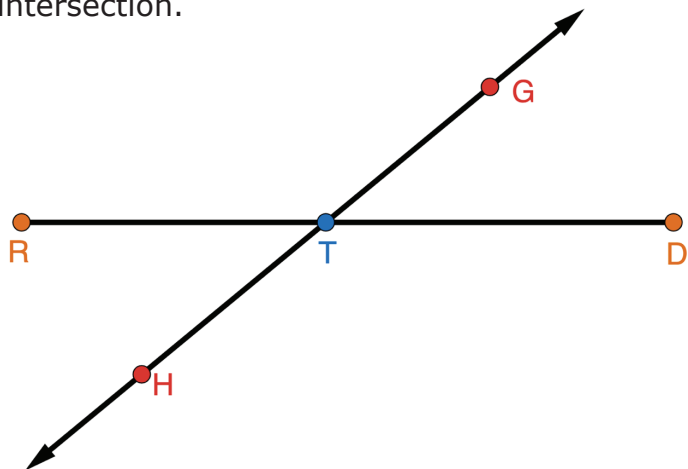
3. $\overline{HL}$ is shown where point N is the midpoint.

- a. Use each description to draw a segment bisector for $\overline{HL}$.



- b. Compare your drawings with your partner.
- c. Using the definition of a segment bisector, discuss with your partner if all of the segment bisectors have to be at the same angle with $\overline{HL}$. Summarize your discussion.

4. $\overline{RD}$ and $\overleftrightarrow{HG}$ are shown where point T is the point of intersection.



- a. Use patty paper to copy $\overline{RD}$.
- b. Fold the copy of $\overline{RD}$ to determine if point T is a midpoint.
- c. Complete the sentence.
 $\overleftrightarrow{HG}$ ☐ is ☐ is not a segment bisector because point T
☐ is ☐ is not a midpoint.



1.10.3 Guided Instruction: Determining Lengths

1. $\overleftrightarrow{KH}$ bisects $\overline{GB}$ at point K .
 - a. Draw a diagram of the description.
 - b. Determine the value of x if $GK = 7x + 36$ and $KB = 13x - 18$.
 - c. What is the length of $\overline{GB}$?

2. $\overline{YH}$ and $\overline{MV}$ bisect each other at point W .

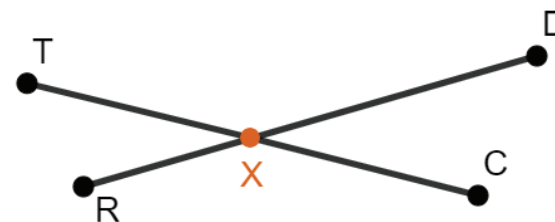
a. Draw a diagram of the description.

b. Determine the value of b if $YH = 12b - 4.5$, $MV = 17b + 0.75$, and $MW - WH = 13.25$.



1.10.4 Guided Practice: Determining Lengths

1. $\overline{RD}$ bisects $\overline{TC}$ at point X .



a. Determine the value of x if $TX = \frac{16}{3}x - 41$ and $XC = 3x + 22$.

b. What is the length of $\overline{TC}$?



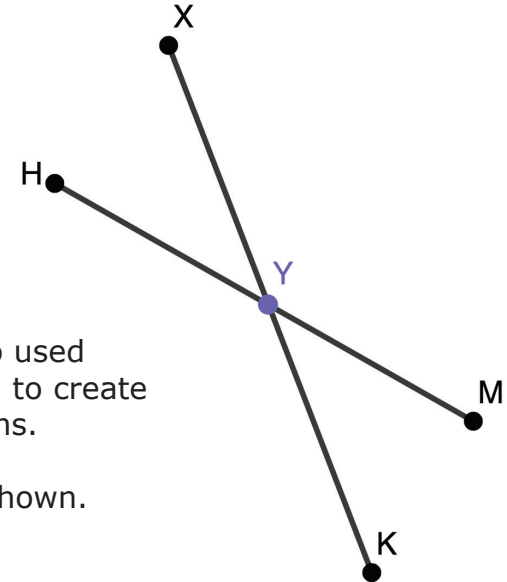
1.10.5 Your Turn!

1. $\overleftrightarrow{BL}$ bisects $\overline{HY}$ at point C .
 - a. Draw a diagram of the description.
 - b. Determine the value of x if $HC = 5x + 9$ and $CY = 8x - 36$.
 - c. What is the length of $\overline{HY}$?



1.10.6 Wrap-Up: Error Analysis

1. $\overline{XK}$ and $\overline{HM}$ bisect each other at point Y , where $XY = 5x + 3$, $HY = 2y + 5$, $XK = 11x + 2$, and $YM = 3y - 2$.



Taliyah and Giancarlo used the given information to create two different equations.

Their equations are shown.

Taliyah	Giancarlo
$5x + 3 = 2y + 5$	$5x + 3 = 2(11x + 2)$

Both Taliyah and Giancarlo made a mistake. If Taliyah was trying to find the value of y and Giancarlo was trying to find the value of x , determine each of their errors. Explain each error.

Taliyah	Giancarlo